Friction: Test the Surfaces · Staff Answers

Learning goal: I can compare friction on different run-out surfaces using repeated measurements and a controlled setup.

Arrival Answers

1. Friction.
2. A shorter distance.
3. Centimetres (cm).
4. Yes. You can ask or pass.

Starter & Hinge Answers

Starter: A. To test the surface, only the run-out surface should change while block, ramp, and release stay controlled.

Hinge: A. Keeping the block, ramp height, and release identical isolates the effect of the run-out surface.

Evidence Desk Cards

B: 24 cm & 26 cm vs A: 46 cm & 48 cm: Supports greater friction on B: shorter stopping distance in both trials.

Same release mark on ramp: Control variable: keeps the speed at the bottom of the ramp consistent.

Different push each time: Flawed method: changes the starting speed so you cannot judge the surface.

Measure to front edge in cm: Control variable: keeps the measurement rule identical.

Accidental bump on Trial 2: Keep and flag: record the 11 cm, note the ruler bump, and take a retry.

Swapping to a heavier block: Flawed method: changing the block adds a second variable.

Check & Repair Answer

Scientists never invent or erase results to match a prediction; surprising results are part of the evidence.

Exit Ticket Answers

1. 24 cm (shorter stopping distance). 2. Same block, same ramp height, same release point, no push. 3. It is part of the evidence; we can flag uncertainty, check the method, and repeat honestly.